Model & Data Card

ForceCRM | Local profile discovery for Greek foodservice businesses

Inputs and intended use

A sales representative uploads customer VAT numbers and revenue. ForceCRM groups customer venues, describes statistically supported traits in Greek and ranks one prospective venue per non-customer VAT. A person verifies suitability before contacting a business.

The supplied GEMI extract contains 24,424 venues, 23,945 distinct VATs and 3,736 Greek activity descriptions. GPS coverage is 96.0%. All 300 portfolio VATs matched. The competition scope covers businesses established during 2020-2024 with a primary activity in division 56.

Profiles in the supplied portfolio

Profile	Customers	Revenue share	Stability[1]	Fit confidence[2]
ICP-01	126	25.8%	93.0%	55.5%
ICP-02	74	27.4%	88.4%	57.8%
ICP-03	100	46.8%	91.4%	68.3%

[1] Mean membership Jaccard across 20 VAT bootstraps with matched profiles and fixed cluster count. [2] Mean relative fit to the profile radius. Sales conversion requires separate outcome data.

Internal retrieval evaluation

Rank cutoff	ForceCRM recall	NACE baseline	Difference
Top 500	25.0%	1.7%	+23.3 percentage points
Top 2,000	55.0%	8.3%	+46.7 percentage points

Seed 2026 held out all venues of 60 customers before fitting on the other 240. The baseline ranks primary-code frequencies in the training portfolio. The organizer retains its separate hidden evaluation. A single development split gives a limited estimate of future sales performance.

Grounding, oversight and data rights

Fisher tests compare trait prevalence at VAT level. Global Benjamini-Hochberg correction, at least five supporting VATs and lift of at least 1.2 limit chance findings. Clustering on the same data makes these associations exploratory. Missing GPS, self-declared activities and multiple branches can distort similarity.

GEMI competition data. Geocoding and venue nodes © OpenStreetMap contributors (ODbL), via Geofabrik. ForceCRM divided ELSTAT 2024 hotel nights by 2021 census residents. ELSTAT bears no responsibility for this transformation. Sales research only.

Model selection and cost

Measurements on the local Intel Core i3-1215U CPU with 7.7 GiB installed RAM

Greek activity embeddings

Multilingual E5-small encodes activity descriptions into 384 dimensions. The US-based Microsoft Research E5 team trained the model on multilingual web and translation pairs, then supervised retrieval data. The benchmark encoded 75 Greek foodservice descriptions and checked top-five retrieval against their four-digit activity families.

Embedding model	Family hit@5	ms / description	Process RSS[3]
e5-small	93.2%	59.5	804 MiB
e5-base	93.2%	195.5	1066 MiB
multilingual-minilm	89.0%	50.3	827 MiB

[3] Sampled after encoding and inclusive of Python allocations. E5-small matched E5-base on this code-family proxy and required less time and memory. Expert judgments would provide a direct test of semantic relevance.

Local Greek narrative drafting

Mistral AI in France developed Ministral 3 3B Instruct. The Q4_K_M package occupies approximately 3 GB. Full corpus provenance remains undisclosed. The LLM selects evidence IDs and Greek sentence variants under a JSON schema. Each displayed sentence links to its source statistic. Generated descriptions carry an AI label.

LLM	Valid drafts / attempted	Generation time[4]
ministral-3:3b	3 / 3	176.4 s
qwen3:1.7b	3 / 3	30.1 s
gemma3:1b	3 / 3	19.3 s

[4] Each model receives the same profiles and evidence. Times are means over valid drafts and retain the original generation latency for cached output. The sample measures schema completion and grounding under the laptop load at run time. Human Greek fluency ratings remain uncollected.

Ministral generated all three final descriptions in 157-182 seconds per profile. Ollama reported 2.34 GiB allocated during CPU inference. The full run took 9.1 minutes with cached activity embeddings. Temperature 0, seed 42, top-k 1 and context 2,048 are fixed. Each export records model identity and source hashes.

Licenses and reproducibility

The distributed Ministral license is Apache 2.0 and allows commercial use under its terms. The E5-small weight repository declares MIT in its metadata and lacks a standalone license file. The official E5 code repository supplies the MIT license. The weight-specific licensing evidence is limited to the publisher's declaration.

OpenAI Codex assisted implementation, documentation and testing. The repository includes source links, an EU AI Act self-assessment, benchmark commands and the temporary Colab workflow.